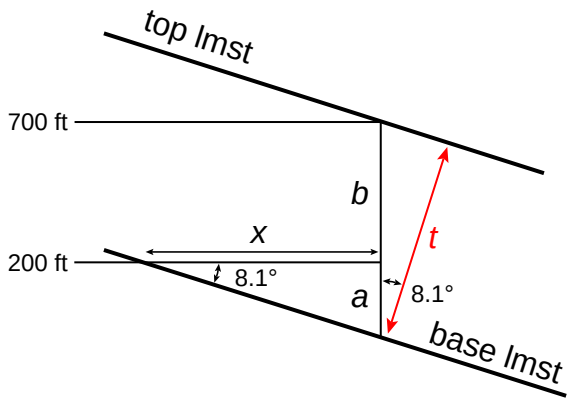
Thickness of limestone

thickness of lmst may be measured from section or calculated using the sketch shown left (not to scale)



x = map distance between 200 ft base of lmst contour (blue) and 700 ft top of lmst contour (green) = 633 ft

$a = x \tan 8.1° = 90.25$ ft $b = 500$ ft

$t = (a + b) \cos 8.1° = 584$ ft

- Strike lines for conglomerate/limestone and limestone/mudstone contacts (blue and green respectively) indicate that these three units all dip 030/8SE (=8/120)
- Dips: dolerite dyke 015/24W (= 24/285) ; ore vein 170/V (= 90°); fault 105/V (= 90°)
- Conglomerate, limestone, mudstone, dyke and vein are all offset by fault
- Strike lines for unconformity (magenta) are not affected by fault (same spacing north and south of fault) and indicate that the unconformity dips 095/3N (=3/005)

Sequence of Geological Events

(1) Deposition of conglomerates, limestones and mudstones in that order

(2) Intrusion of dolerite dyke [*crosscuts conglomerates, limestones and mudstones*]

(3) Tilting [*located here assuming that the vein and fault were originally vertical; if they were not, then the tilting is only constrained to after (1) but before (6)*]

(4) Intrusion of ore vein [*crosscuts dolerite dyke*]

(5) Faulting [*offsets ore vein*]

(6) Uplift and erosion to form unconformity [*not offset by fault*]

(7) Deposition of sandstones [*above unconformity*]

(8) Tilting [*sandstones are gently tilted*]

(9) Erosion to present topography